IE301 · TOPIC 1 · FINAL PRESENTATION

Two times per patient

Cutting waiting and overtime in an Indian day-care chemotherapy unit

Team INFUCINQ
Parav · Sahasra · Siddhesh · Vaibhav · Kartikya

316 → 157 min

mean wait before the drip

127 → 10 min

overtime after closing, per day

0

extra chairs, nurses or doctors needed

THE PROBLEM

Patients wait 3–4 hours before the drip starts

1	2	3	4	5
Registration	**Blood draw**	**Lab result**	**Doctor review**	**Billing**
5–15 min	5–10 min	60–120 min	10–20 min	30–60 min
6	**7**	**8**	**9**	**10**
Drug prepared	**Wait for chair**	**IV line, pre-meds**	**Infusion**	**Discharge**
15–30 min	20–60 min	20–40 min	30 min – 6 h	15–30 min

Orange steps are done the same morning by most Indian patients, and the drug cannot be mixed until the doctor clears the blood counts. Flow confirmed by oncologists at Tata Memorial, 1 Oct 2026.

3.5 h

mean wait before chemo at Amrita, Kochi (1,029 patients)

300–400

patients a day on 70–80 chairs at Tata Memorial

5 h 40

average time in the unit at an Indian centre

THE PROBLEM

The binding constraint is ready time, not chairs

37%

of chair-hours sit empty while the patient is already in the unit but not yet ready

Simulated, everyone told 08:00, 30% pre-cleared.
Presentation 3 predicted 35% from the stage times alone.

Why

- A same-day patient told 08:00 is ready for a chair only at about 11:10: registration, blood, 90 min lab, review, billing and drug.
- Until then only pre-cleared patients can use the chairs, so mornings are empty and afternoons run late.
- Calling everyone at 08:00 adds a crowd: 88 min queue at registration alone.

What the literature leaves open

Hesaraki 2019, 2020 chair and nurse template; ready time known in advance

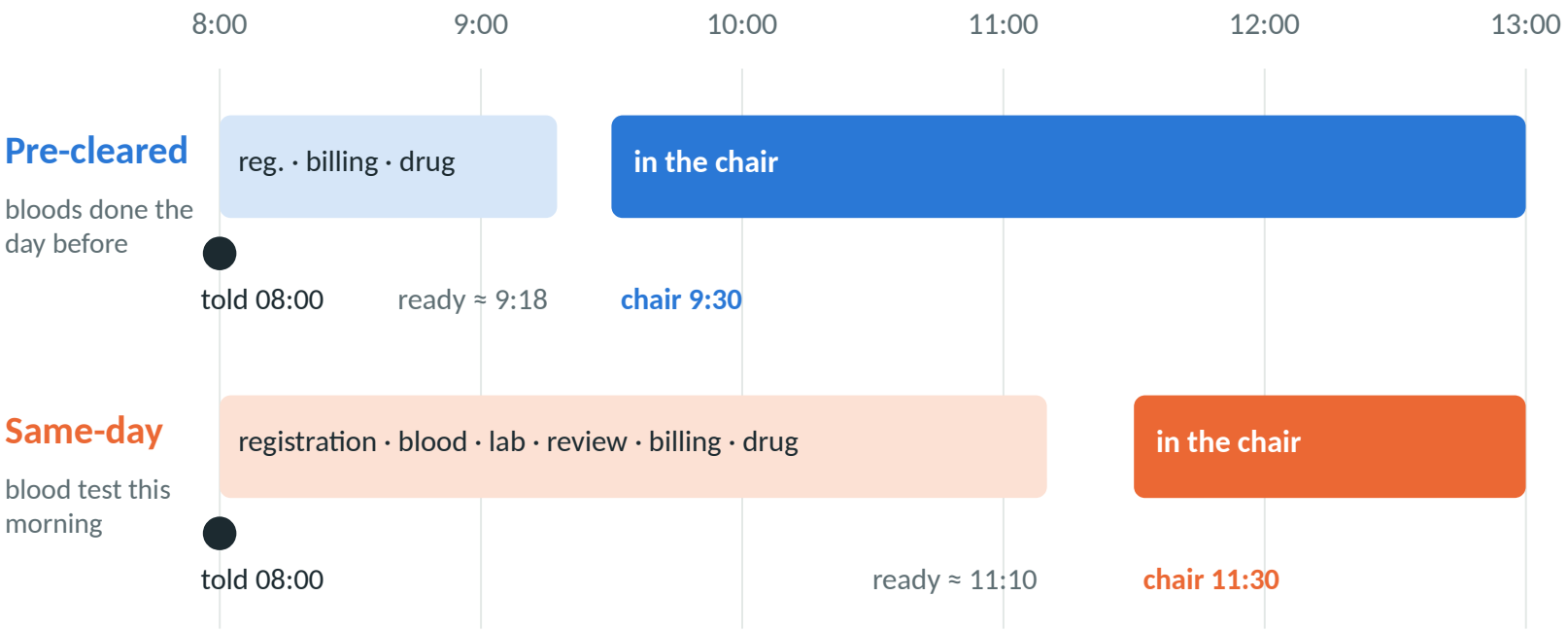
Liang 2015 optimise the schedule, then stress-test it in simulation

Hadid 2022 every stage random; 13 h to optimise 246 appointments

Open: setting the appointment from an uncertain ready time, and varying how many patients are pre-cleared.

OUR PROPOSAL

Give every patient two times: arrival and chair



Bars show average times from our Presentation 3 inputs. Booking a same-day patient's chair at 08:30 would leave it empty for about 2 h 40.

The rule

chair slot = arrival + B

B covers 80% of patients of that class, so 4 in 5 are ready in time.

90 min

pre-cleared buffer (89.9 min)

210 min

same-day buffer (206.2 min)

OUR PROPOSAL

Three changes, no new chairs or staff

1

Two-time booking

An integer programme sets every chair slot inside the chair and nurse limits; arrival = slot − B.

Overtime 127 → 37 min E1

2

Pre-clear 60% the day before

Phone triage and outside labs, as Kochi did; their drugs are prepared before opening.

Overtime 37 → 10 min E3, E5

3

Cap arrivals per 15 min

Never send more patients than registration (3) and the doctors (2) can see in 15 minutes.

Front-desk queues halved E7

DATA

No public data exists, so we built a calibrated dataset

5,007
patients

100
simulated days

22
inputs, each with a source

23 / 23
data checks passed

Input	Value used	Source
Registration · doctor review	Tri(5, 10, 15) · Tri(10, 15, 20) min	Presentation 3; TMH visit
Blood draw + lab result	Tri(5, 7.5, 10) + Tri(60, 90, 120) min	Presentation 3; TMH 4 h tested
Billing · drug preparation	Tri(30, 45, 60) · Tri(15, 22.5, 30) min	Presentation 3; EJHP 2017
Line and pre-meds (1 nurse)	Tri(20, 30, 40) min	Hesaraki et al. 2019
Infusion length mix	30 min 45% · 2 h 18% · 3 h 18% · 4 h 11%	Clinic day, n = 55; Liang 2015
Actual vs planned infusion	short +17%, long −6%	Hacettepe study, Bath, Mayo
Unfit at review · no-show	8% · 5%	NCT00883181 · assumed
Arrival vs told time	Tri(−30, 0, 30) min	Presentation 3 placeholder

Kaggle and the literature hold no per-patient infusion-unit timestamps. Every stage is drawn per patient from a fixed seed, so every rule is tested on the same 100 days. Our buffers reproduce Presentation 3 exactly: 89.9 and 206.2 min.

SIMULATION

A discrete-event simulation of the 10-step flow

Registration
2 desks

Blood · lab · review
same-day only · 2 doctors

Billing
approval delay

Pharmacy
3 hoods

Chain + nurse
20 chairs · 8 nurses

Infusion → home
1 nurse watches 4

Nurse rule (Hesaraki 2019): 4 × (setups + takedowns) + patients on the drip ≤ 4 × nurses, at every minute.

Built three ways: Python SimPy (all results), an interactive web simulator, and an AnyLogic build kit that replays the same days.

10 / 10 code checks

hand calculation, capacity never exceeded, patients conserved, Little's law

Check	Published	Model
Kochi wait	3.5 h	**4.1 h**
Time in unit	5 h 40	**6 h 28**
Morning chairs lost	35%	**37%**
Buffers (min)	89.9 / 206.2	**exact**

FORMULATION

Formulation: a buffer rule plus an integer programme

1 Booking rule

$$P(a_i + D_i \le s_i) \ge q$$
$$\Rightarrow\ s_i = a_i + B,\quad B = q\text{-quantile of } D$$

a = told arrival, s = chair slot, D = random delay to drug ready, B set per class.

q = 0.8 gives B = 90 and 210 min.

Newsvendor view: best q = idle cost ÷ (idle cost + wait cost).

Two numbers carry all the randomness, so the daily problem stays deterministic and fast.

2 Daily chair slots (time-indexed MIP)

y(g, t) = patients of group g starting at 15-min slot t; z(τ) = unit still open at τ

$$\min\ \Sigma\ \text{finish time} + w_m \cdot \text{day length} + w_o \cdot \text{overtime}$$

Everyone booked	$\Sigma_t\, y(g, t) = n(g)$ for every group
Chairs	patients in chairs at $\tau \le 20 \cdot z(\tau)$
Nurses	$4 \cdot$ (setups + takedowns) + drips $\le 4 \cdot N$
Buffer	$y(g, t) = 0$ before open + B
Open is one block	$z(\tau) \ge z(\tau + 1)$
Arrival cap	told per 15 min ≤ 3, same-day ≤ 2

FORMULATION

The MIP finds the true best schedule in 7 seconds

Test case	Schedules tried	Brute force	MIP	
1	4,800	378	378	exact
2	12,000	208	208	exact
3	2,560	infeasible	infeasible	agree
4	2,880	infeasible	infeasible	agree
5	1,152	infeasible	infeasible	agree
6	4,800	384	384	exact

Brute force: every combination of start times for 4 patients, 2 chairs, 2 nurses. The MIP matches all 6, including proving 3 cases have no feasible schedule.

48%

better objective than the greedy list schedule, every one of 20 real days

11 min

planned overtime, against 76 min for greedy

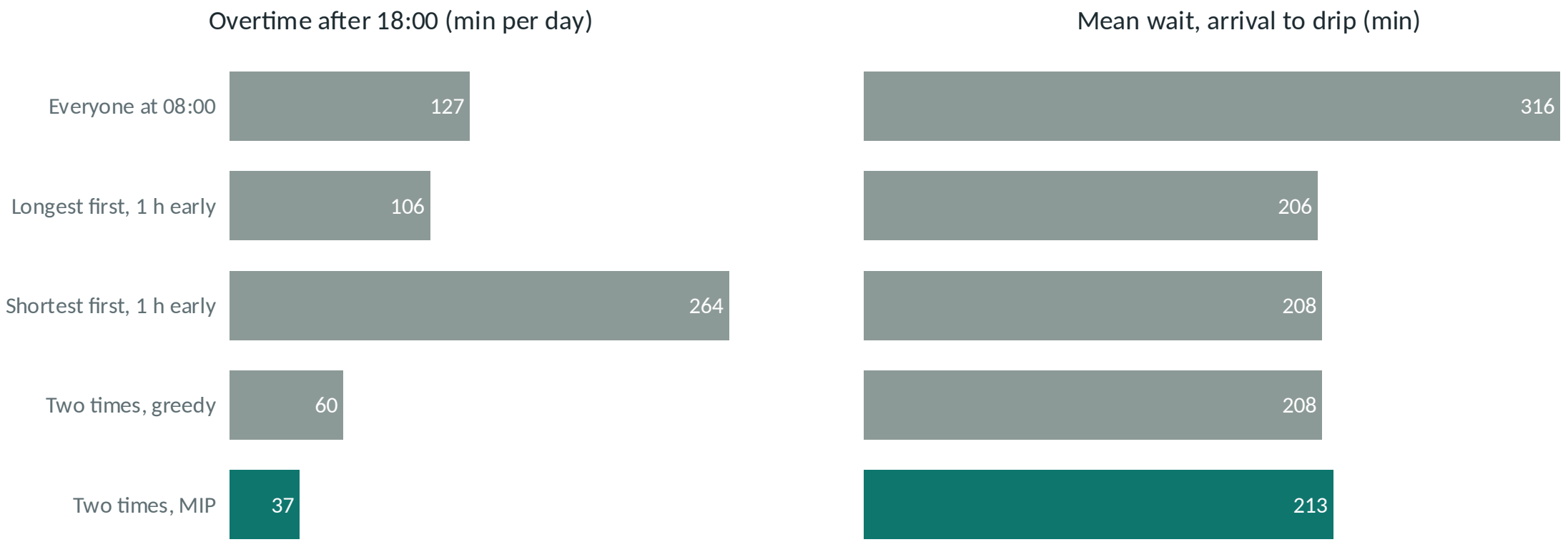
6.9 s

mean solve time per day (560 variables)

13 h

for 246 appointments in Hadid 2022's full search

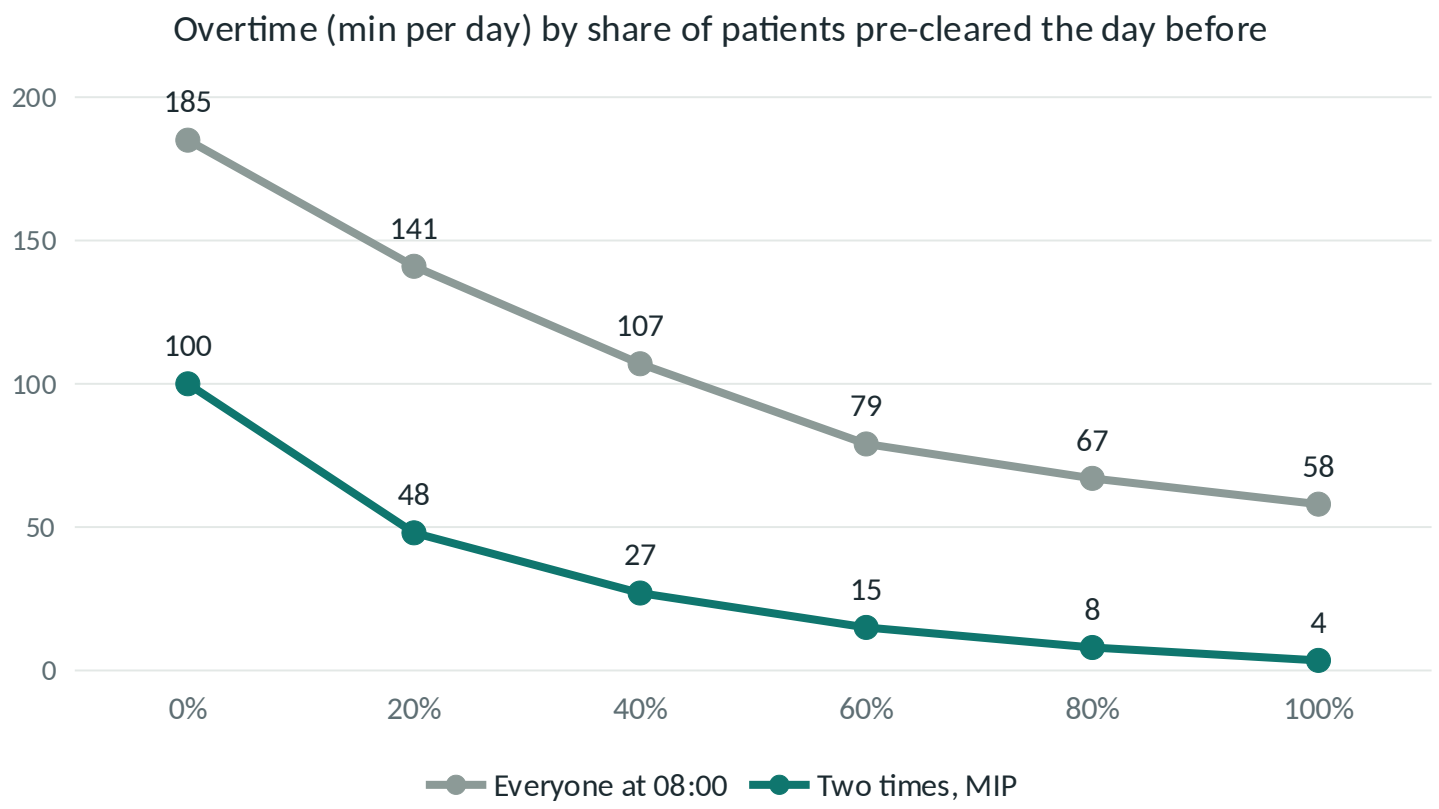
Two-time booking cuts overtime by 71%



100 identical simulated days, 30% pre-cleared, 20 chairs, 8 nurses. 95% CIs: overtime ±5–13 min, wait ±2–4 min. Against everyone-at-08:00: overtime −71%, wait −33%.

RESULT 2 · PRE-CLEARANCE

Pre-clearing 60% of patients captures most of the gain



At 60% pre-cleared, two-time booking

15 min

overtime (100 at 0%)

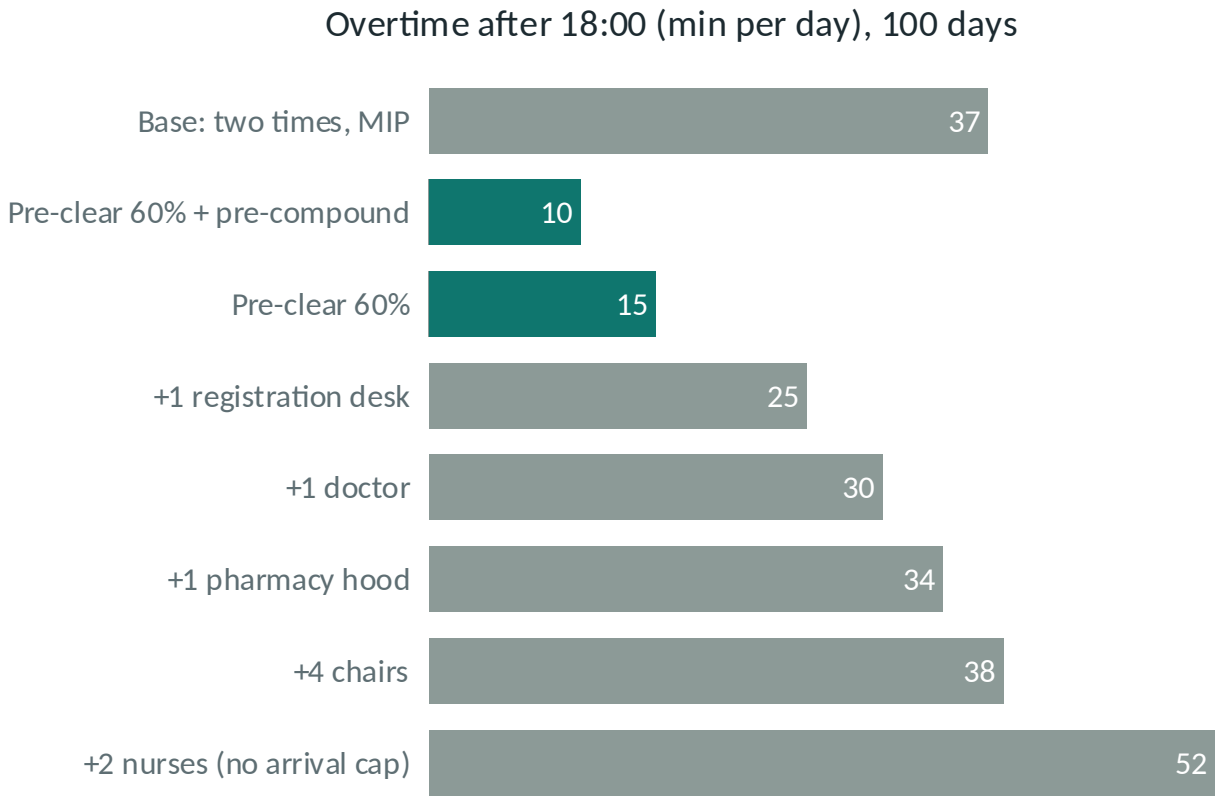
178 min

mean wait (238 at 0%)

Beyond 60%, overtime falls only 15 → 4 min. Waiting keeps falling to 125 min, because each pre-cleared patient skips about 2 h of tests.

Under everyone-at-08:00, wait stops improving after 60% (271 → 250 min): the morning crowd at registration dominates, so pre-clearance needs two-time booking alongside it.

Process changes beat buying capacity



Chairs are not the constraint

+4 chairs: 37 → 38 min. No change.

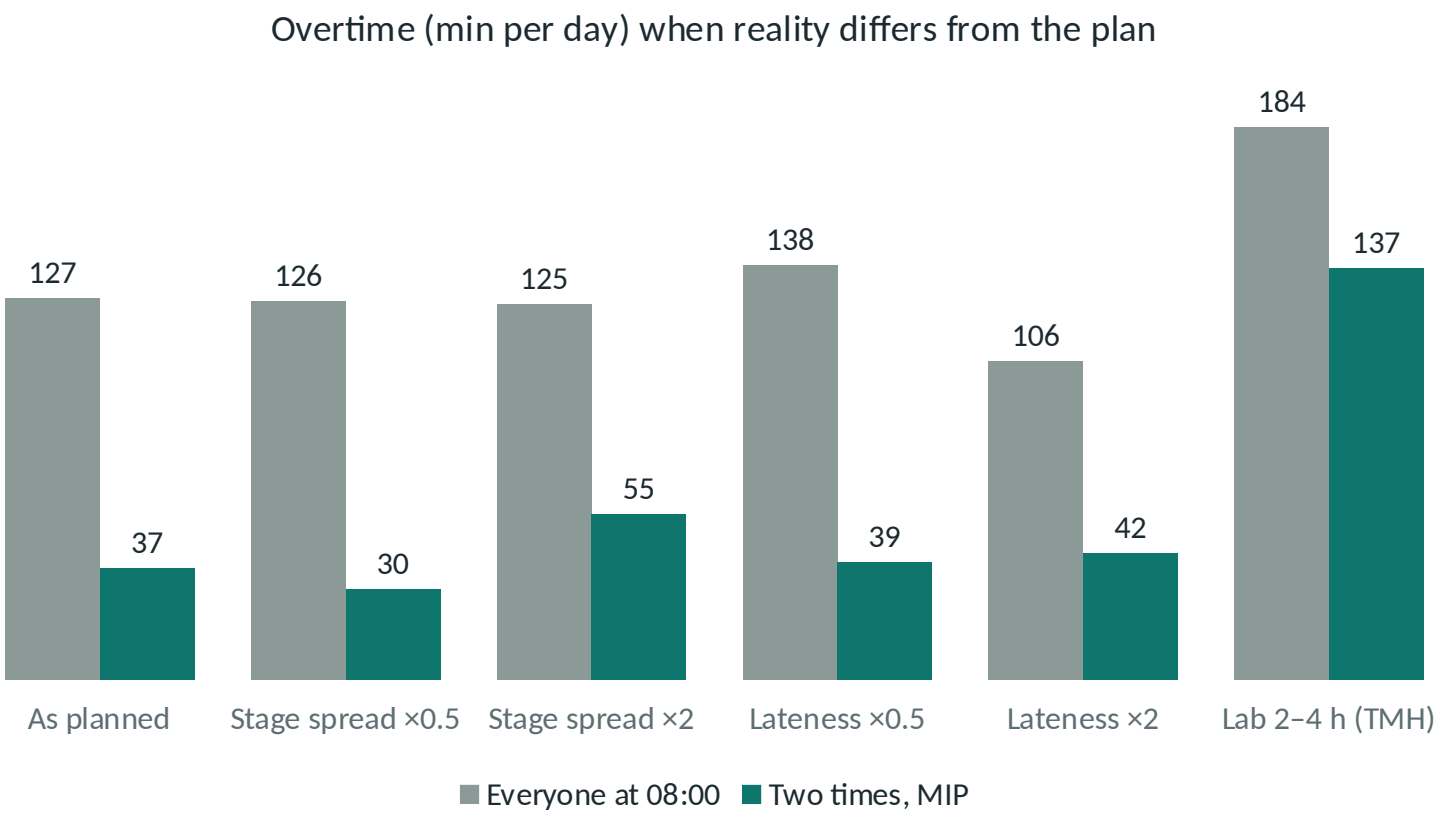
Extra nurses moved the queue

Without the arrival cap, the plan bunched arrivals: registration queue 13 → 28 min. With the cap, 10 nurses cut overtime to 28 min.

Best package, no new staff

Pre-clear 60% and pre-compound: wait 157 min, overtime 10 min. With the arrival cap: 145 min (40-day run).

The plan survives wrong inputs, except a slow lab



Two-time booking wins in all 6 cases

Doubling every spread costs 18 min of overtime (37 → 55).

A 2–4 h lab breaks the buffers: 137 min overtime. Measure lab turnaround first at TMH.

WHAT WE ACHIEVED

Same staff: half the wait, a tenth of the overtime

−50%

mean wait before the drip

316 → 157 min

−92%

overtime after closing

127 → 10 min a day

−71%

overtime from booking alone

127 → 37 min

0

new chairs, nurses or doctors

process changes only

Two-time booking + 60% pre-cleared + pre-compounded drugs, against everyone told 08:00. 100 identical simulated days, 20 chairs, 8 nurses.

ASSUMPTIONS AND LIMITS

What these numbers rest on

Data

- Synthetic, calibrated: stage times are the Presentation 3 ranges, not yet measured at TMH
- Published figures taken from abstracts; full texts still to check
- Infusion mix from one 55-patient clinic day

Unit

- Generic unit: 20 chairs, 8 nurses, 08:00–18:00, 60% of chair-hours booked
- Presentation 3's 80% load is infeasible: 156 min overtime
- TMH runs 24 h with walk-ins; walk-ins are outside the model

Behaviour

- A ready patient may start before the slot if a chair and nurse are free
- Setup and takedown each need one whole nurse; one nurse watches 4 drips
- Billing is an approval delay, not a queue; unfit 8%, no-show 5%

Model

- The MIP plans with mean durations; the simulation adds all randomness
- 15 s solver limit, 1% gap; greedy fallback if no solution
- Buffers sized for a 60–120 min lab

Summary

Problem Patients wait 3–4 h because same-day blood work makes them ready only around 11:00; 37% of chair-hours are lost to it.

Proposal Two times per patient, arrival and chair, with chair = arrival + an 80% buffer (90 / 210 min), slots set by an integer programme.

Evidence Calibrated dataset (23/23 checks), verified simulation (10/10), MIP matches brute force (6/6), 100 identical days per scenario.

Result Booking alone: overtime −71%. With 60% pre-cleared and pre-compounding: wait 316 → 157 min, overtime 127 → 10 min.

Next Measure TMH stage times, lab turnaround first, and refit every input.

Telling each patient two times, and clearing 60% of them the day before, halves the wait and nearly removes overtime, with no new chairs or staff.